

## 11.8 Correlation

In [Chapter 5](#) we already introduced the **stochastic** concept of **correlation**:

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}}$$

The **statistical** concept of **correlation** is defined in an analogous way. The estimator of  $\rho$  is:

$$\begin{aligned} R(X, Y) &= \frac{\text{cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}} \\ &= \frac{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}) \cdot (Y_i - \bar{Y})}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \cdot \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}} \end{aligned}$$

After reduction:

$$R(X, Y) = \frac{\sum_{i=1}^n X_i Y_i - n \cdot \bar{X} \cdot \bar{Y}}{\sqrt{(\sum_{i=1}^n X_i^2 - n \cdot \bar{X}^2) \cdot (\sum_{i=1}^n Y_i^2 - n \cdot \bar{Y}^2)}}$$

We call  $R(X, Y)$  the **sample correlation coefficient**.

The correlation coefficient  $R(X, Y)$  has the following features:

- $R(X, Y)$  is a measure of **linear relationship**;
- $-1 \leq R(X, Y) \leq 1$ ;
- if  $R(X, Y) = \pm 1$ , then there is a **complete** linear relationship, i.e. the scatter plot forms a straight line;
- if  $R(X, Y) = 0$ , then there is **no** linear relationship (i.e. either a non-linear relationship or no relationship whatsoever).

### Example 3

See the data from Example 1: the relationship between *outside temperature* and *energy consumption*.

$$\begin{aligned} r(X, Y) &= \frac{193990 - 12 \cdot \frac{685}{12} \cdot \frac{3325}{12}}{\sqrt{\left(43245 - 12 \cdot \left(\frac{685}{12}\right)^2\right) \cdot \left(927875 - 12 \cdot \left(\frac{3325}{12}\right)^2\right)}} \\ &= 0.8025 \end{aligned}$$

So there is a **strong positive linear relationship** between *outside temperature* and *energy consumption*.

### Remarks

- It is also possible to determine  $r(X,Y)$  using the option **LinRegTTest** of the **graphical calculator**.

Check:

$$r = 0.8025.$$

- The **LinRegTTest** option also examines whether the correlation with size 0.8025 found in the **sample** indicates a correlation present in the **population**.

The following test is carried out:

$$\begin{array}{ll} H_0: & \rho = 0 \quad \text{"no correlation"} \\ H_1: & \rho \neq 0 \quad \text{"correlation"} \end{array}$$

If  $H_0$  is rejected, then the correlation coefficient is called **significant**.

In particular:

$$\begin{array}{ll} r = 0.8025^* & \text{for } 0.01 < \alpha \leq 0.05 \\ r = 0.8025^{**} & \text{for } \alpha \leq 0.01 \end{array}$$

Since  $p = 0.0017$  it applies:

$$r = 0.8025^{**}.$$

- The appropriate **test statistic** for the **LinRegTTest** is:

$$T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$$

If  $H_0: \rho = 0$  is true,  $T$  follows a  $t$  distribution with  $n - 2$  degrees of freedom.

Check:

$$\begin{aligned} t &= \frac{0.8025 \cdot \sqrt{12-2}}{\sqrt{1-0.8025^2}} \\ &= 4.2539 \end{aligned}$$

Since  $t = 4.2539 > C_R = \text{invT}(.995, 10) = 3.1693$ , we reject  $H_0$  at  $\alpha = 0.01$ . So  $r = 0.8025^{**}$ .

Using a  $P$ -value:  $p = \text{tcdf}(4.2539, 10^{\wedge}10, 10) = 0.0017 \leq 0.01$ . So reject  $H_0$  at  $\alpha = 0.01$ .

### Simple Linear Regression and Correlation using Excel

Fill columns A and B with the data from Example 1. Choose an empty cell (e.g. D2) to calculate the **intercept**. Next, choose *Insert Function* (i.e. the first button left to the *Formula Bar*), select a category ("Statistical") and a function (INTERCEPT). Finish with OK.

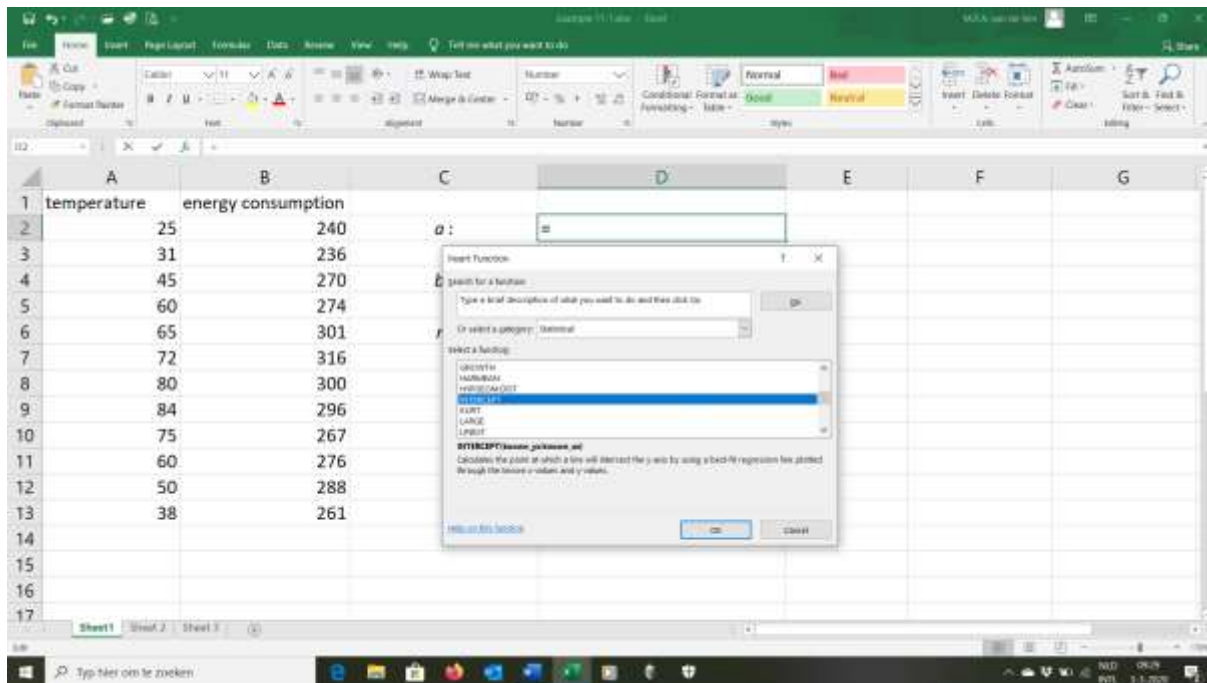


Figure 11.4 Determining the intercept using Excel (1)

Select  $y$ 's (i.e. B2:B13) and select  $x$ 's (i.e. A2:A13). Finish with OK.

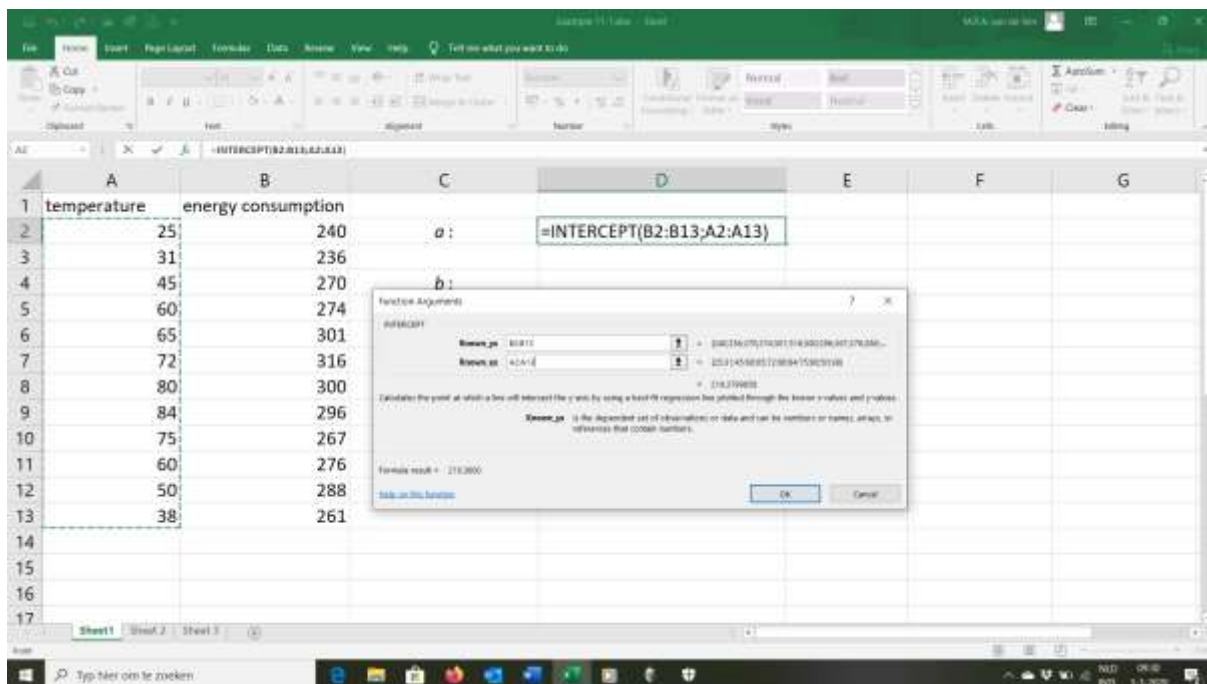


Figure 11.5 Determining the intercept using Excel (2)

So the **intercept** is 219.3800.

Repeat this procedure for  $b$  (the **SLOPE** function; cell D4) and  $r$  (the **CORREL** function; cell D6).

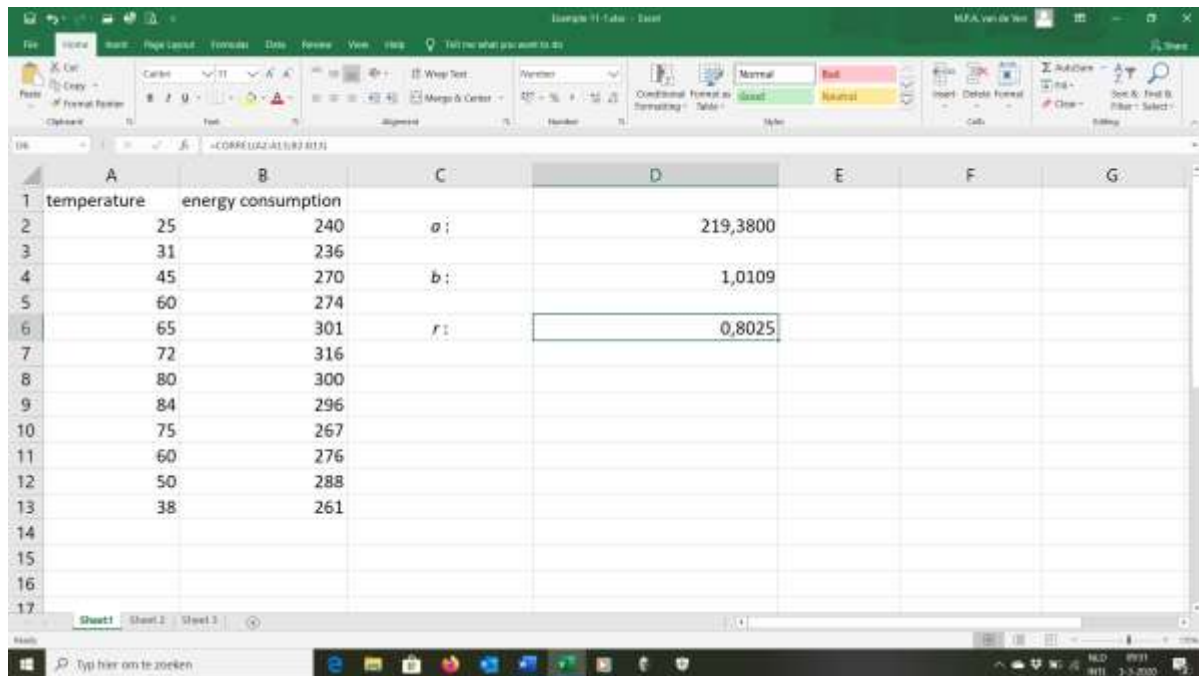


Figure 1.6 Determining the slope and the correlation coefficient using Excel

So the **slope** is 1.0109. The **correlation coefficient** is 0.8025.

Finally, it is also possible to use Excel to calculate all relevant quantities at the same time. For this, it is necessary to enable two **add-ins**, namely *Analysis ToolPak* and *Analysis ToolPak – VBA*. Choose *File – Options – Add-ins*. Select *Analysis ToolPak*, click *Go* and tick *Analysis ToolPak* and *Analysis ToolPak – VBA*.

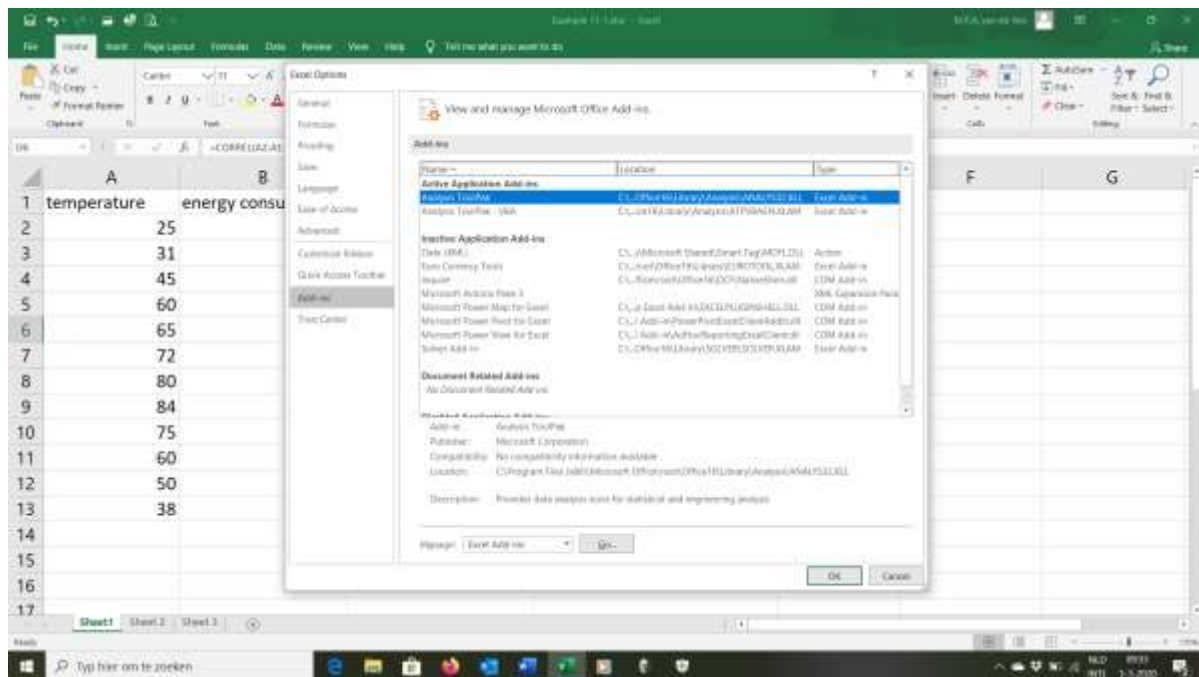


Figure 1.7 Activating Applications Add-inns

Go to the *Data* tab and choose **Data Analysis** (i.e. the rightmost button). Select **Regression**.

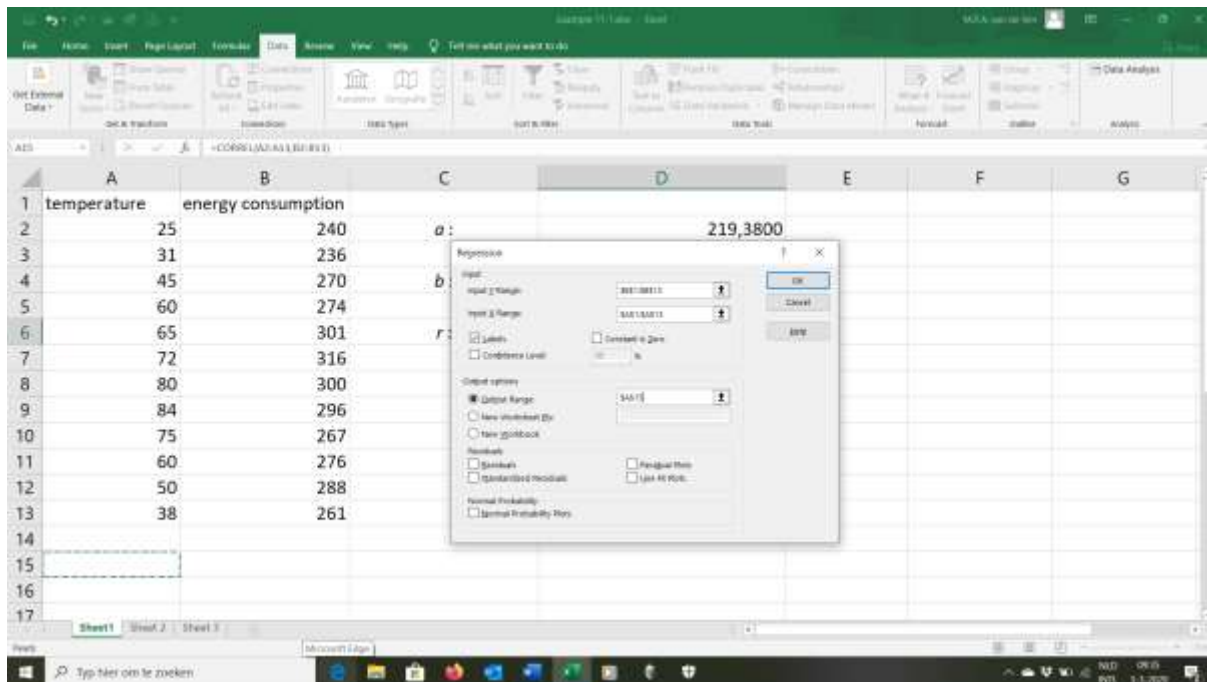


Figure 1.8 Determining relevant regression quantities using Excel (1)

Finish with OK.

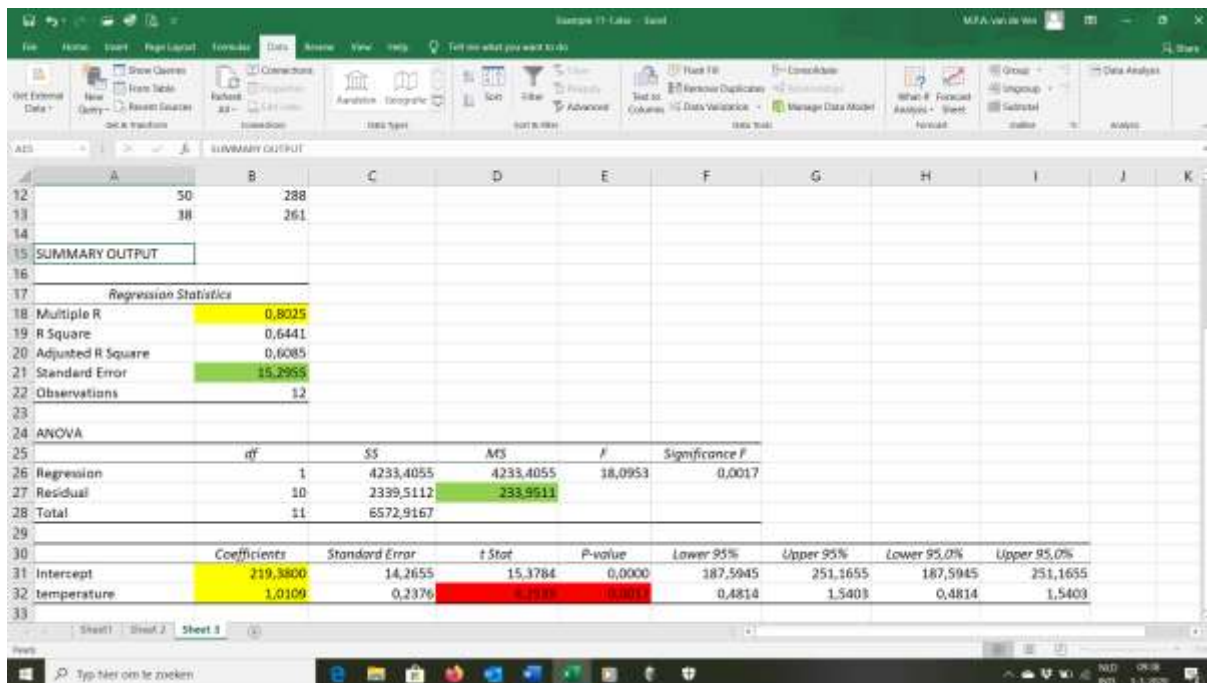


Figure 1.8 Determining relevant regression quantities using Excel (2)

In yellow:  $r$ ,  $a$  and  $b$ . In green: estimates of  $\sigma$  and  $\sigma^2$ . In red: the test statistic  $t$  and its  $P$ -value.